THE PICARD STACK IS SMOOTH OVER THE BASE FOR RELATIVE DIMENSION AT MOST 1

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1582-1587 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 1589-1619, the complete author proof (29 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 9 of queue_moduli.json (pb=1537, stmt 1582-1587, proof 1589-1619).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DMB: sources/stacks-moduli/pinned-original-source.tex, source lines 1526–1529. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A4T: sources/stacks-moduli/pinned-original-source.tex, source lines 1619–1619. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08EX: sources/stacks-moduli/pinned-original-source.tex, source lines 1619–1619. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09ZX: sources/stacks-moduli/pinned-original-source.tex, source lines 1619–1619. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DNM: sources/stacks-moduli/pinned-original-source.tex, source lines 1619–1619. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DP0: sources/stacks-moduli/pinned-original-source.tex, source lines 1619–1619. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Assume $f : X \to B$ has relative dimension ≤ 1 in addition to the other assumptions in this section. Then $\mathcal{P}ic_{X/B} \to B$ is smooth.*

Proof. We already know that $\mathcal{P}ic_{X/B} \to B$ is locally of finite presentation, see Lemma 0DMB. Thus it suffices to show that $\mathcal{P}ic_{X/B} \to B$ is formally smooth, see More on Morphisms of Stacks, Lemma 0DP0. Taking base change, this immediately reduces to the following problem: given a first order thickening $T \subset T'$ of affine schemes, given $X' \to T'$ proper, flat, of finite presentation and of relative dimension ≤ 1, and for $X = T \times_{T'} X'$ given an invertible $\mathcal{O}_X$-module $\mathcal{L}$, prove that there exists an invertible $\mathcal{O}_{X'}$-module $\mathcal{L}'$ whose restriction to X is $\mathcal{L}$. Since $T \subset T'$ is a first order thickening, the same is true for $X \subset X'$, see More on Morphisms of Spaces, Lemma 09ZX. By More on Morphisms of Spaces, Lemma 0DNM we see that it suffices to show $H^2(X, \mathcal{I}) = 0$ where $\mathcal{I}$ is the quasi-coherent ideal cutting out X in X'. Denote $f : X \to T$ the structure morphism. By Cohomology of Spaces, Lemma 0A4T we see that $R^p f_* \mathcal{I} = 0$ for $p > 1$. Hence we get the desired vanishing by Cohomology of Spaces, Lemma 08EX (here we finally use that T is affine). □